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المدرسة الوطنية العليا للذكاء الاصطناعي
قسم هندسة الأنظمة الذكية

Human Computer Interaction Labs 4th Year **2026/2027**

Lab 3 (II) Semester: S7

User-Centered Design Applications - part II
Heuristics in User-Centered Design

In Lab 3 Part I, we explored *User-Centered Design (UCD)* principles and their application in real-world cases such as *Airbnb*. Now, we will connect those ideas to the concept of **heuristics**. While UCD focuses on involving users throughout the design process, heuristics provide **expert-based evaluation rules** that help identify issues without direct user testing.

In Lab 2, you worked with **Design Rules** (principles like consistency, feedback, and affordance). While those are general design guidelines, **Heuristics** (Such as: *Nielsen's*) are a specific, widely-used set of *usability rules*. There are also other heuristic frameworks (e.g., *Gerhardt-Powals' cognitive engineering principles* or *Shneiderman's Eight Golden Rules*), which differ slightly in focus, emphasizing cognitive load, while others emphasize interface consistency or task efficiency.

This lab will compare different design evaluation methods and deepen your understanding of UCD.

Objectives:

- Understand and compare different heuristic frameworks (Nielsen, Shneiderman, Gerhardt-Powals, etc.).
- Apply heuristics to evaluate real-world systems and prototypes.
- Reflect on the connection between UCD principles, personas, and heuristic evaluation.
- Understand the impact of AI on testing the heuristics.
- **Work in groups of 3–4 to encourage collaborative reflection and richer findings, and stimulate creativity**.

Pre-Lab [30 mins Discussion]

Reading & Reflection: Work in groups, review these frameworks and the articles, write down your answers to be presented during the lab session **(No LLMs)**:

A. **Notions:**

1. Nielsen's 10 Usability Heuristics
2. Shneiderman's Eight Golden Rules
3. Gerhardt-Powals' Cognitive Engineering Principle.

Reflection Questions:

- Present similarities/differences between frameworks.
- Each group highlights *one real-life example from Airbnb (or another app)* where a heuristic applies/fails. **Why?**
- Connect these insights back to UCD principles from earlier labs or an app you use daily.

B. **Articles:**

1. Heuristics Considering UX and Quality Criteria for Heuristics
2. Ten User Experience Heuristics

Reflection Questions:

- What new heuristics (beyond those in Part A) did you discover in these papers?
- Do you think some heuristics are more universal (apply to any system) while others are context-dependent (e.g., mobile, AI, VR)? Give examples.
- Can heuristics sometimes conflict (e.g., minimalist design vs. showing enough help)? How would you resolve that?

Lab Work

Activity 1: Cross-Heuristic Challenge [20 mins]

This activity should be done in groups and will conclude with an open discussion with the whole class..

A. Each group chooses a popular app or website (Airbnb, Spotify, Amazon, Duolingo, AliExpress - Shopping App, Temu: Shop Like a Billionaire. . .). Identify at least **3 examples where the system satisfies heuristics** and **3 example where it fails**.

B. Each group picks one heuristic framework (Nielsen, Shneiderman, Gerhardt-Powals, SMASH. . .). Using the Airbnb case study (**Lab 3- Part I**), evaluate the website/app through the lens of that framework. Which framework gave more practical insights? Which felt harder/easier to apply?

Activity 2: Heuristic Remix: Design Challenge [25 mins]

Combine your knowledge of the concept of heuristics, to:

A. Redesign one small part from one of your old projects. Use Figma Prototype, or just pen and paper.

B. **OR**, Redesign one small part of Airbnb (e.g., search bar, booking confirmation, host profile,...): Figma 1, Figma 2.

C. Apply at least **3 different heuristics** from any framework to justify your redesign.

Activity 3: AI & Creativity with Heuristic Testing [15 mins]

A. **Prototype Testing:** Use a simple prototype or UI image (Your old projects, AirBnB,...).

B. **AI Exploration:** Try out at least one free AI tool to support heuristic evaluation. Suggested tools:

- Use multiple multimodal LLM (ChatGPT, Gemini, Claude,...) for an AI-powered heuristic evaluation for websites/UIs. You can upload a screenshot.
- Explore SynthUX (tool exploration) AI-powered UX audit in your browser. Explore its features and evaluation workflow. **Note that performing a complete heuristic evaluation using this tool may take considerable time. Therefore, you are not required to conduct a full analysis during the lab session. You may explore the tool independently at home to understand how it works**.

C. **Task:** Compare the AI feedback with Nielsen's heuristics. Which heuristics are easy to detect with AI, and which require human judgment?

D. **Class Reflection:** How AI can accelerate heuristic evaluation? Where AI fails and human expertise is necessary?